

Customer Value Mapped by Purchasing Behavior - US Candy Distributor

Cover Page	Background	RFM Scores - How They Were Developed	Customer Retention Status - How This Wa..	Customer Score Distribution	Regional Popularity and Sales	Product Preference..
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Customer Profiles and Retention for a US Can..



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Background

US Candy Company is a division of a larger U.S.-based confectionery retailer. Known for its niche, high-quality chocolates, the company has cultivated a loyal customer base but now aims to strengthen its presence in the U.S. market by both deepening relationships with existing customers and attracting new buyers.

To guide their marketing and product development strategies, senior executives want a clearer understanding of their current customers - who they are, w..

Customer profiling has proven to be a valuable strategy for many industries. According to various case studies, businesses that implement effective customer profiling can achieve:

- More targeted and efficient marketing campaigns

..

Data and Approach

The analysis is based on sales data from 5,044 unique customers, covering the period January 2021 through December 2024. The objective is to build comprehensive customer profiles that provide actionable insights into customer demographics and purchasing behavior.

Key areas of focus include:

- Location
- Preferred products
- Purchase frequency and spending behavior..

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RFM Scores - How They Were Developed

For the purposes of marketing and customer base expansion, a US Candy Distributor wants to look into the "who" and "how" of the buyers currently making up it's customer base. To a..

Elements of a Customer Score:

Recency: Defines how recently a customer purchased from a company (for example, 25 ..

Frequency: This is the frequency of orders placed by the customers within the given per..

Monetary: This is the sum of the Customer's cumulative Gross Profit.

$$\begin{array}{ccccc} \textit{Last Order Date - Recency} & & \textit{Count (\# of Unique Order IDs} & & \textit{Sum (Gross Profit for each Customer} \\ \textit{Parameter} & + & \textit{associated with a single Customer} & + & \textit{Order)} \\ & & \textit{ID)} & & \end{array}$$

These values are percentile ranked amongst the other values in their columns, ranked from high to low - and then normalized to adjust to a standardized metric.

Recency Score + Frequency Score + Monetary Score

Customer Score:

..

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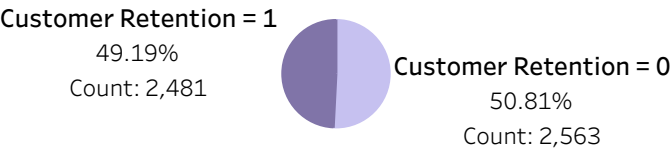
Customer Retention

Customer Retention:

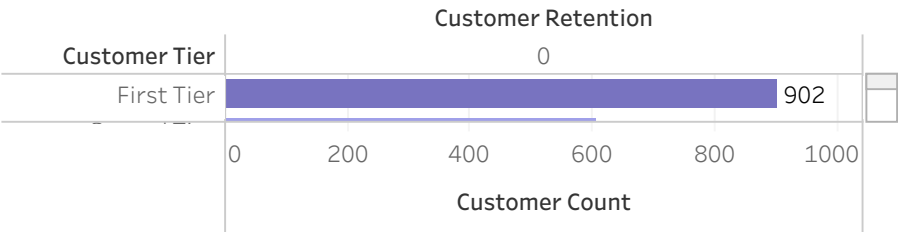
This was the value of whether a customer became a repeat purchaser of the Candy Distributors’ or not (1 = True / Retained, 0 = False / Not Retained). This value was determined based o..

Customer Retention Count

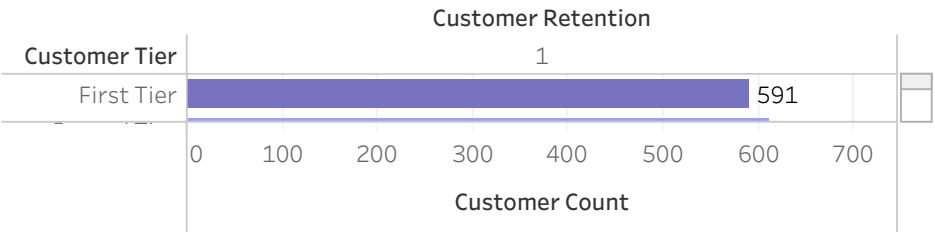
(1 = Retained, 0 = Not Retained)



Customer Retention = 0:



Customer Retention = 1:



We can see that First Tier Customers make up a large proportion of those that are Retained.

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Background	RFM Scores - How They Were Developed	Customer Retention Status - How This Wa..	Customer Score Distribution	Regional Popularity and Sales	Product Preference Across Customer Tier..	Seasonality
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Customer Score Distribution..

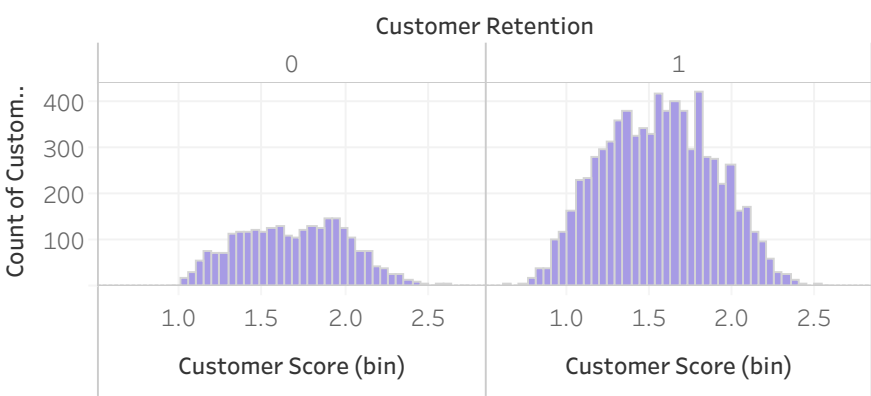
The Customer Score value holds a fairly normal distribution. This means customer behavior is relatively similar across tiers. Since we see a fairly normal dis..

Mean	Median	Mode	Range	Standard Deviation ..	Mean..	Mode..	Stan- dard D..	Mean..	Mode..	Stan- dard D..
1.591	1.590	1.78	2.067		Median..	Range..		Median..	Range..	

Customer Score Distribution - Raw



Customer Score Distribution by Retention



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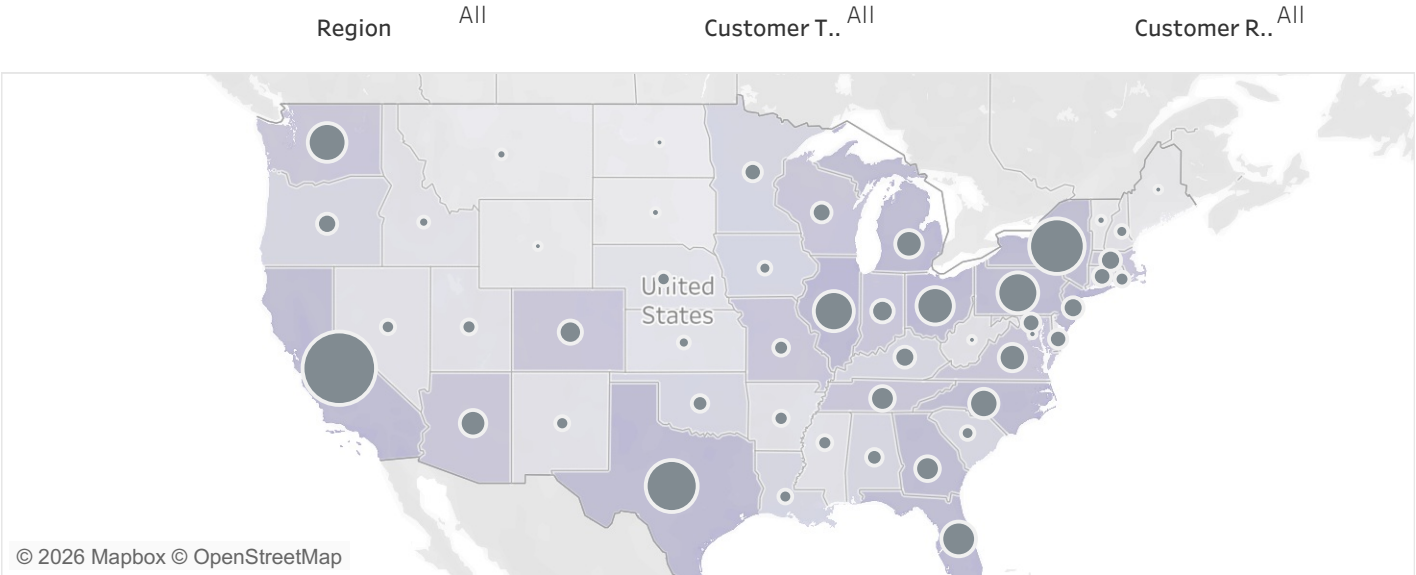
RFM Scores - How They Wer..	Customer Retention Status - How This Wa..	Customer Score Distribution	Regional Popularity and Sales	Product Preference Across Customer Tier..	Seasonality	LTV (Lifetime Value)
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Customer Count and Percent Contribution to Profit ..

Buyers are fairly spread out throughout the states, with concentrated enclaves of customer presence, and with that, profits. The top states for this US Can..

Regional Profits (in 2024)

Region	Gross Profit	Contribution to Profit
Pacific	\$10,550	
Atlantic	\$8,811	
Interior	\$7,035	
Gulf	\$4,555	



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Customer Retention Stat..	Customer Score Distribution	Regional Popularity and Sales	Product Preference Across Customer Tier..	Seasonality	LTV (Lifetime Value)	Recommendations
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Product Preferences

Wonka Bar - Milk Chocolate was the most popular product by order count (1,768), but was outpaced by the Scrumdiddlyumptious Wonka Bar in profits (\$19,358 vs \$17,443). As a group, the Chocolate product category was the most popular among customer tiers. This could be an e..

Customer Retention

- ☒ 0
- ☒ 1

Product Popularity - Sheet

Customer Tier

- ☒ First Tier
- ☒ Second Tier
- ☒ Third Tier
- ☒ Fourth Tier

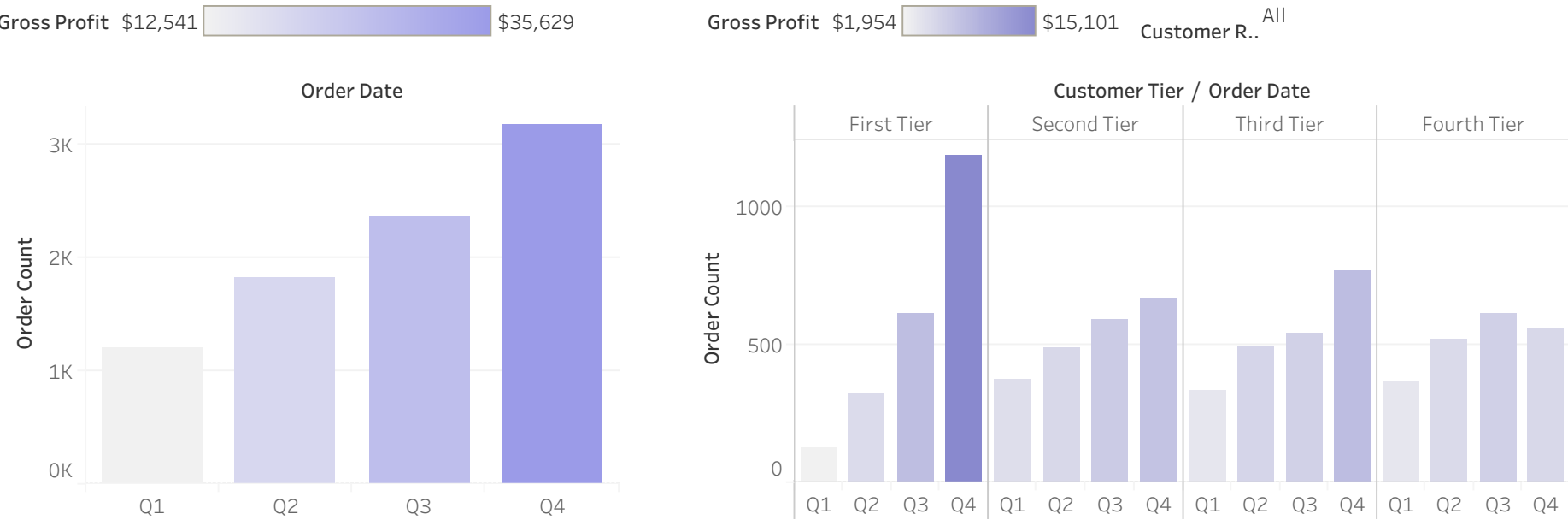


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Customer Retention..	Customer Score Distribution	Regional Popularity and Sales	Product Preference Across Customer Tier..	Seasonality	LTV (Lifetime Value)	Recommendations
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Seasonality: Quarterly Order Count

Ordering peaks in the last quarter of the year, during the winter months. We see that First Tier customers in particular participate in a surge of orders toward the end of the year. Focusing on production and operations end of year, could be helpful in terms of driving up profits. Similarly, specialized marketing campaigns for the winter season and holiday months might also be supportive in encouraging more customers to purchase or in converting more..



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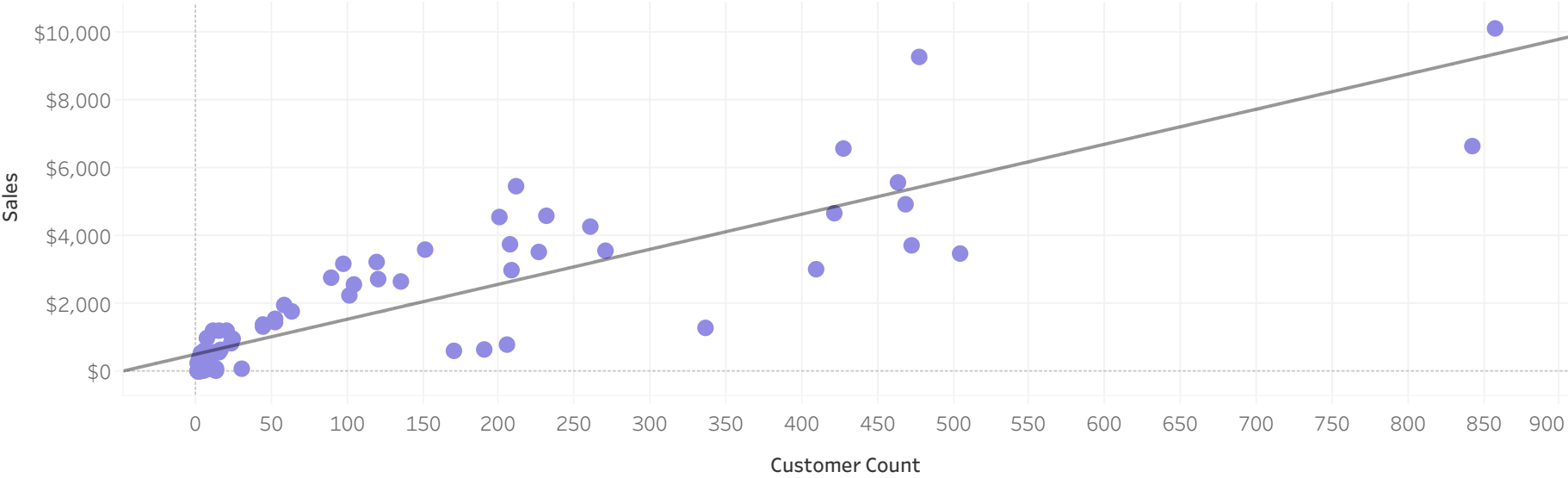
Customer Retention..	Customer Score Distribution	Regional Popularity and Sales	Product Preference Across Customer Tier..	Seasonality	LTV (Lifetime Value)	Recommendations
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Lifetime Value (LTV)

Lifetime Value represents the lifetime revenue earned from a customer. LTV was calculated using the total sales value for each customer for all purchases made over the..

Customer Retention
All

Customer Tier
All



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Customer Retention..	Customer Score Distribution	Regional Popularity and Sales	Product Preference Across Customer Tier..	Seasonality	LTV (Lifetime Value)	Recommendations
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Recommendations..

Regional The Candy Distributor has customer coverage in nearly all 50 states. Regionally, we see a greater concentration of customers as well as profits in a handful of states - California, New York, Texas, Pennsylvania,..

Year of Ord.. 2024

Region	Gross Profit	Contribution to T..
Pacific	\$10,550	34.09%
Atlantic	\$8,841	28.47%

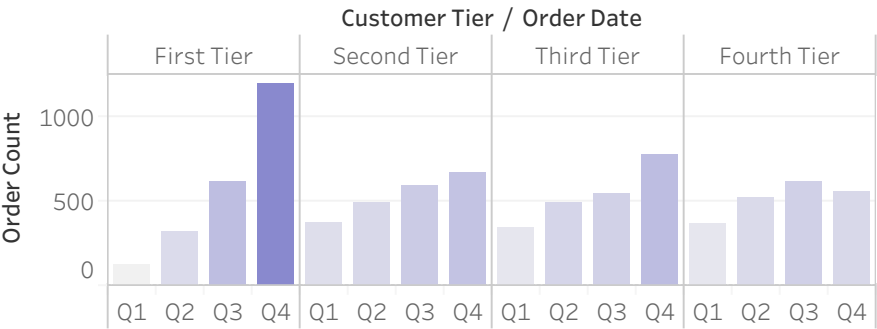
Preferred Products Top Product varied a bit among tiers, but as a common factor - the Chocolate product category performed well among all customers, with the Scrumdiddlyumptious bar being the top performer out of all categories. With this, we would propose that the Di..

Customer T.. All	
Division ..	Product Name
Chocolate	Wonka Bar - Scrum
	Wonka Bar - Triple
	Wonka Bar - Milk C
	Wonka Bar - Nutty

Seasonal In the chart, we can see a peak in ordering in Q4. This pattern tends to cross tiers as well with nearly all tiers seeing a significant increase in order count during the last quarter of the year. Given this, we can recom..

Gross Profit \$1,954 \$15,101

Order Count by Quarter (Tier)



With this, it may be possible to convert a good chunk of the existing customer base into "First Tier" customers - and potentially expand to new buyers.